ROSE5730 Autonomous Mobile Robots

Lecture 3: Kinematics

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Outline

Locomotion

3 Mobile robot maneuverability

2 Kinematic constraints

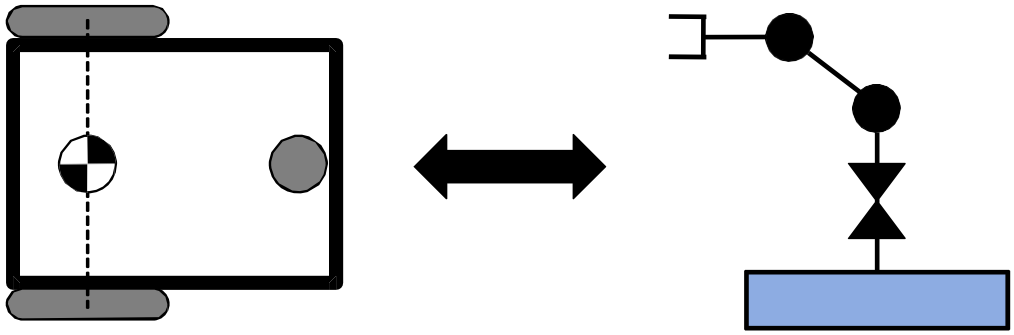
1 Kinematic model



Mobile robot kinematics: Overview

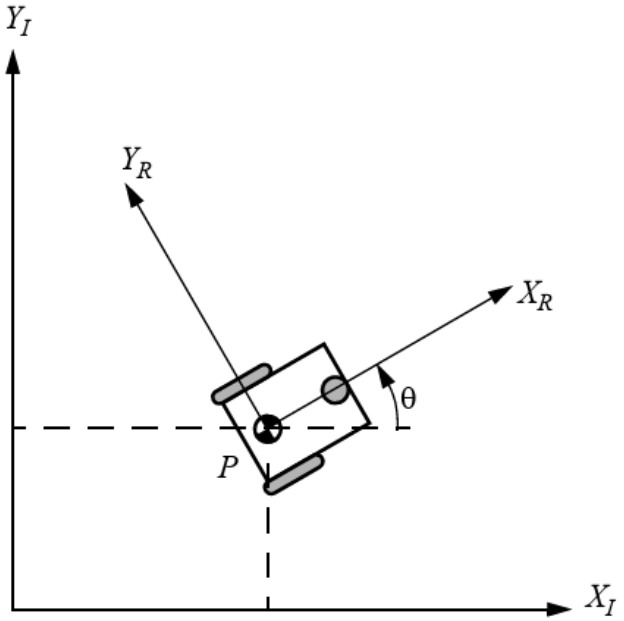
- Manipulator arms versus mobile robots
  - Both are concerned with forward and inverse kinematics. However, for mobile robots, encoder values do not map to unique robot poses
  - mobile robots can move unbound with respect to their environment
  - There is no direct (=instantaneous) way to measure the robot's position
  - Position must be integrated over time, depends on path taken. Leads to inaccuracies of the position (motion) estimate

- Understanding mobile robot motion starts with understanding wheel constraints placed on the robot's mobility



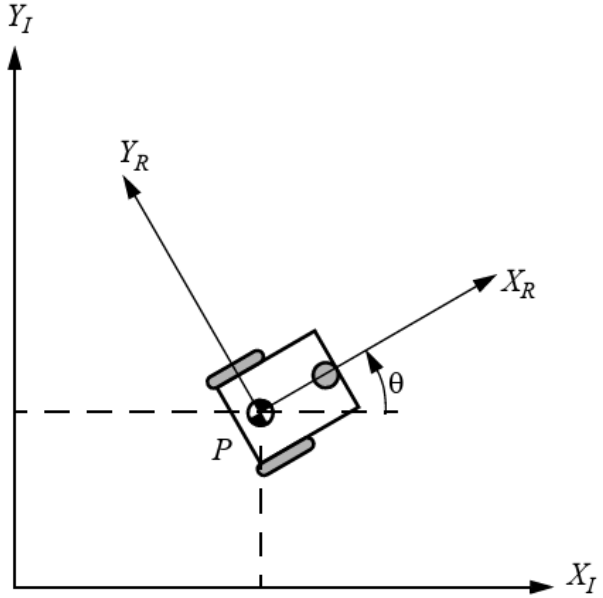
Representing robot position

- The total dimensionality of this robot chassis on the plane is three, two for position in the plane and one for orientation along the vertical axis, which is orthogonal to the plane.
- By robot chassis we refer only to the rigid body of the robot, ignoring the joints and degrees of freedom internal to the robot and its wheels. control expense.



Representing robot position

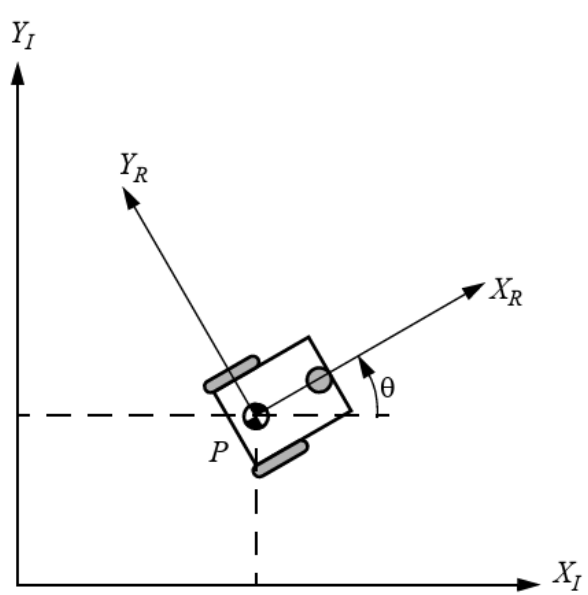
- The axes X_I and Y_I define an arbitrary inertial basis on the plane as the global reference frame from some origin O: $\{X_I, Y_I\}$, and $\{X_R, Y_R\}$ defines two axes relative to P on the robot chassis and is thus the robot's local reference frame.
- The position of P in the global reference frame is specified by coordinates x and y, and the angular difference between the global and local reference frames is given by θ.

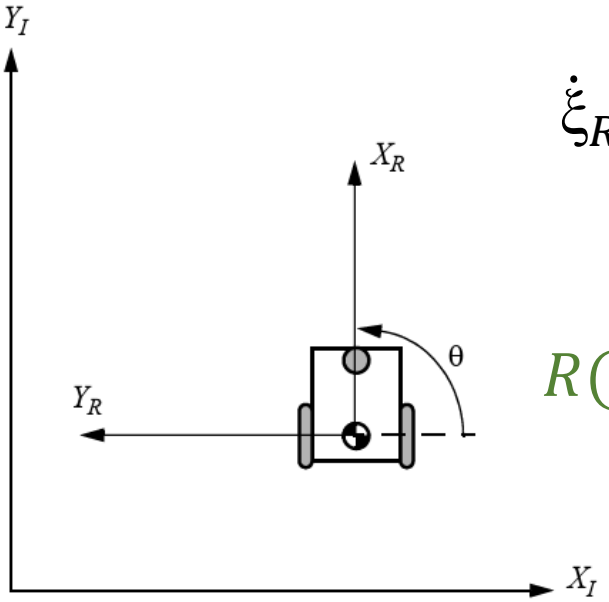


$$\xi_I = \begin{bmatrix} x \\ y \\ \theta \end{bmatrix}$$

Representing robot position

- To describe robot motion in terms of component motions, it will be necessary to map motion along the axes of the global reference frame to motion along the axes of the robot's local reference frame.





$$\dot{\xi}_R = R(\theta)\dot{\xi}_I$$

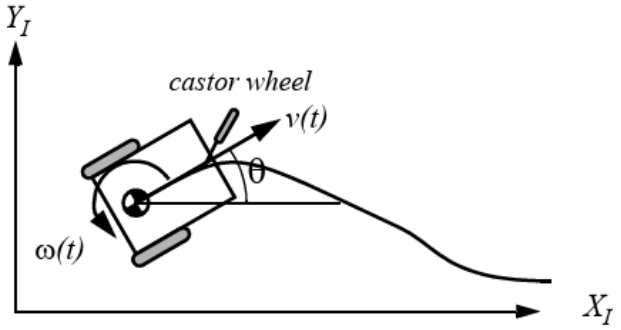
$$R(\theta) = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\theta = \frac{\pi}{2}$$

Forward kinematic model

- Given a point P centered between the two drive wheels, each wheel is a distance l from P. Given r, l, θ, and the spinning speed of each wheel $\dot{\varphi}_1$, and $\dot{\varphi}_2$.
- A forward kinematic model would predict the robot's overall speed in the global reference frame:

$$\dot{\xi}_I = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = f(l, r, \theta, \dot{\varphi}_1, \dot{\varphi}_2)$$





Kinematic model for the differential-drive robot

$$\dot{\xi}_I = \mathrm{R}(\theta)^{-1} \begin{bmatrix} \frac{r\dot{\varphi}_1}{2} + \frac{r\dot{\varphi}_2}{2} \\ 0 \\ \frac{r\dot{\varphi}_1}{2l} + \frac{-r\dot{\varphi}_2}{2l} \end{bmatrix} = \mathrm{R}(\theta)^{-1} \begin{bmatrix} \dot{x}_\mathrm{R} \\ \dot{y}_\mathrm{R} \\ \dot{\theta}_\mathrm{R} \end{bmatrix} \qquad \mathrm{R}(\theta)^{-1} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Example

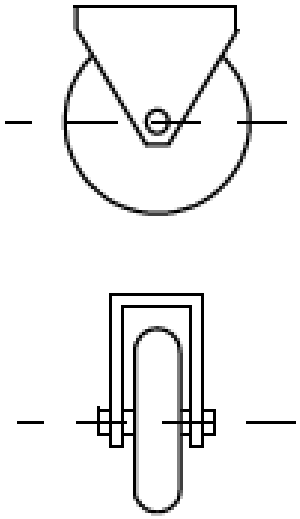
- Suppose that the robot is positioned such that $\theta = \frac{\pi}{2}$, $r = 1$, and $l = 1$. The robot engages its wheels unevenly with the spinning speed of each wheel $\dot{\varphi}_1 = 4$, and $\dot{\varphi}_2 = 2$. Try to compute $\dot{\xi}_I$
- Solution:

$$R(\theta)^{-1} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\dot{\xi}_I = R(\theta)^{-1} \begin{bmatrix} \frac{r\dot{\varphi}_1}{2} + \frac{r\dot{\varphi}_2}{2} \\ 0 \\ \frac{r\dot{\varphi}_1}{2l} + \frac{-r\dot{\varphi}_2}{2l} \end{bmatrix} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \\ 1 \end{bmatrix}$$

Kinematic constraints

- Assumptions:
  1. The plane of the wheel always remains vertical and that there is in all cases **one single point of contact** between the wheel and the ground plane.
  2. There is **no sliding** at this single point of contact. That is, the wheel undergoes motion only under conditions of pure rolling and rotation about the vertical axis through the contact point.

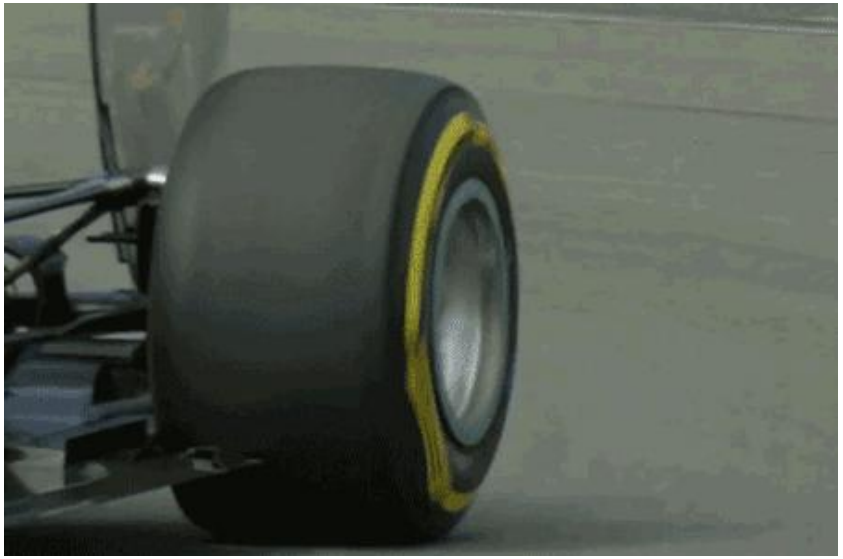




Kinematic constraints

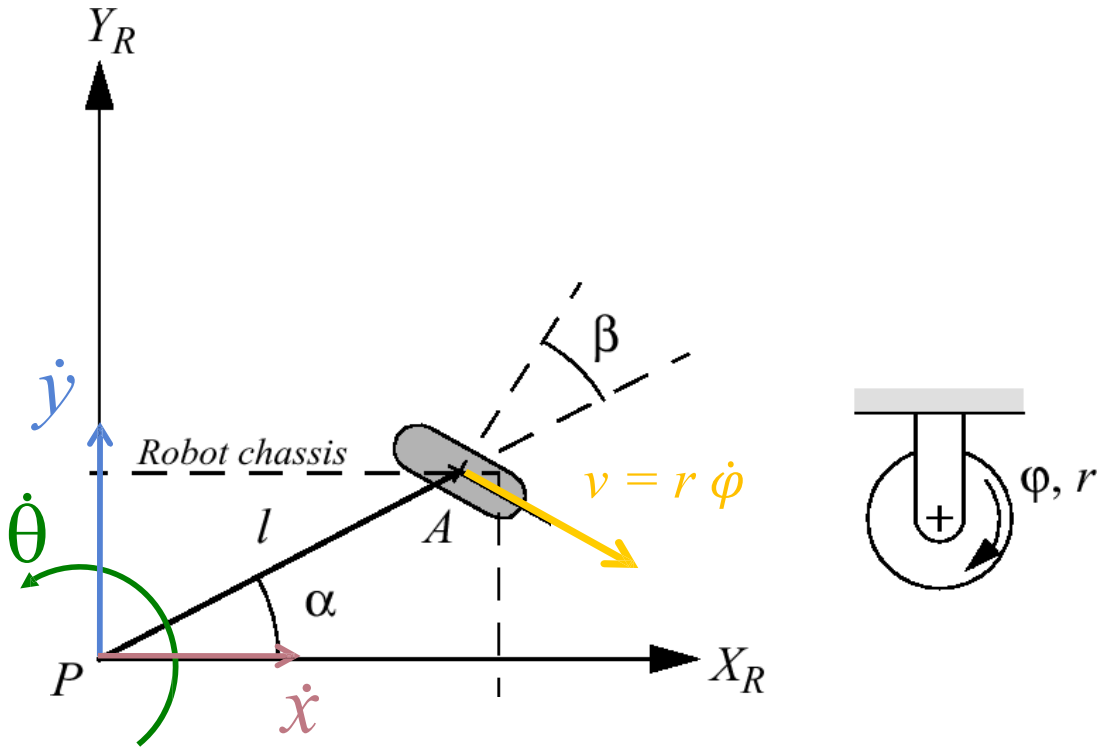
- **The first constraint** enforces the concept of rolling contact: the wheel must roll when motion takes place in the appropriate direction. (No forward/backward slipping)
- **The second constraint** enforces the concept of no lateral slippage: the wheel must not slide orthogonal to the wheel plane. (No lateral slipping)

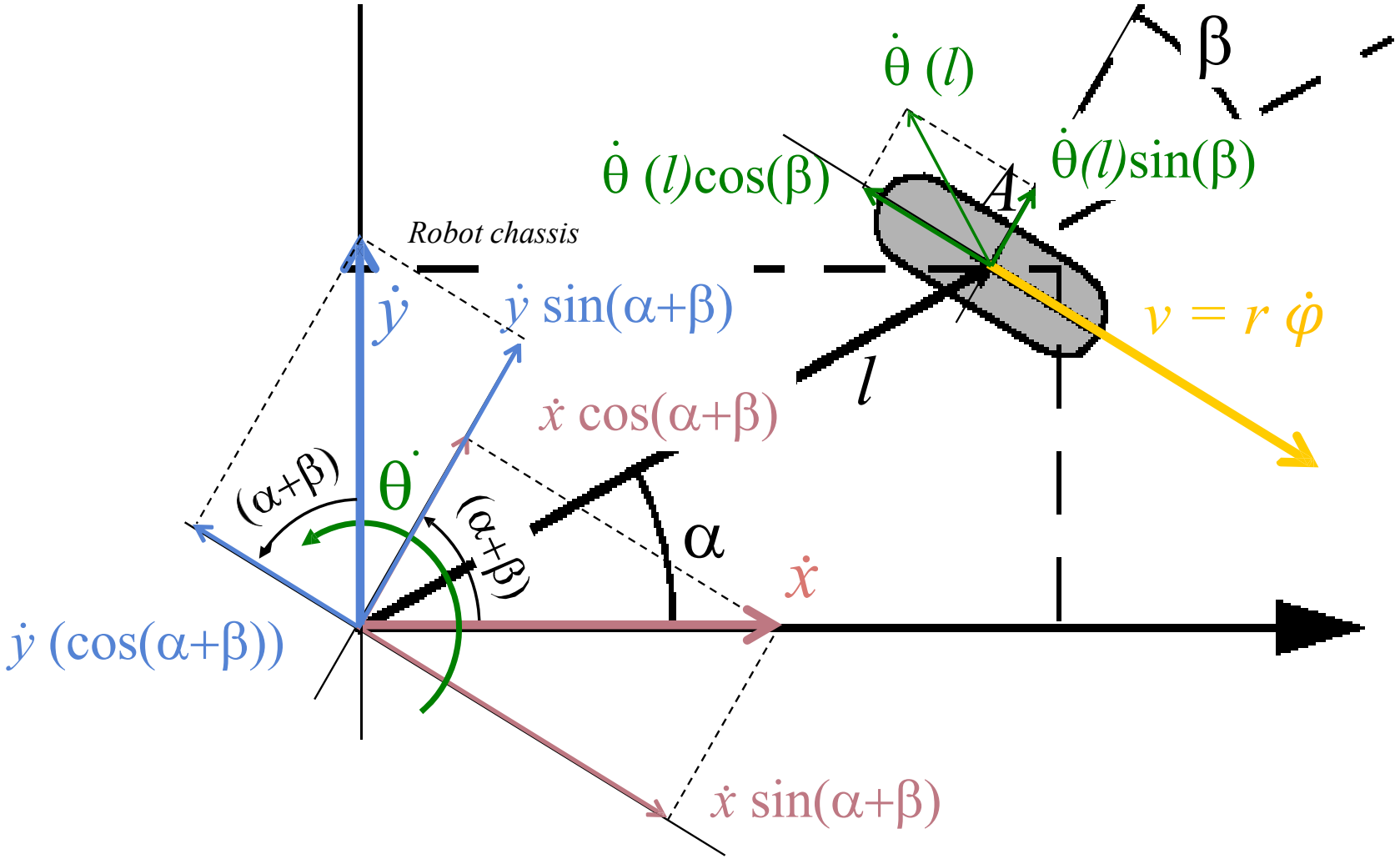




Fixed standard wheel

The position of A is expressed in polar coordinates by distance l and angle α. The angle of the wheel plane relative to the chassis is denoted by β, which is fixed since the fixed standard wheel is not steerable. The wheel, which has radius r, can spin over time, and so its rotational position around its horizontal axle is a function of time t: $\varphi(t)$



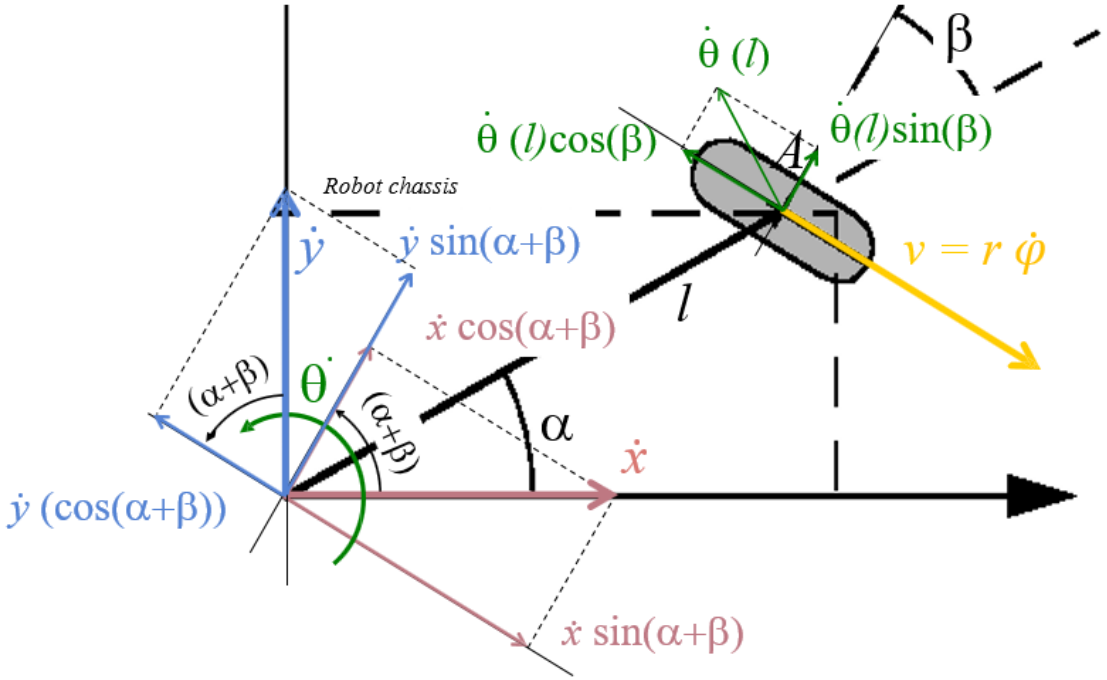

Robot chassis
$\dot{y}$
$\dot{y}\sin(\alpha+\beta)$
$\dot{x}\cos(\alpha+\beta)$
$\dot{\theta}$
$(\alpha+\beta)$
$(\alpha+\beta)$
α
$\dot{x}$
$\dot{y}\,(\cos(\alpha+\beta))$
$\dot{x}\sin(\alpha+\beta)$
$\dot{\theta}\,(l)$
$\dot{\theta}\,(l)\cos(\beta)$
$\dot{\theta}(l)\sin(\beta)$
β
A
l
$v = r\,\dot{\varphi}$

Fixed standard wheel

The rolling constraint: The total motion of car along the direction of wheel plane = Motion of wheel spin.

$$[\sin(\alpha+\beta) \quad -\cos(\alpha+\beta) \quad (-l)\cos\beta\,]\; \mathrm{R}(\theta)\dot{\xi}_I - r\dot{\varphi} = 0$$

$$\dot{\xi}_R = \begin{bmatrix} \dot{x}_R \\ \dot{y}_R \\ \dot{\theta}_R \end{bmatrix}$$

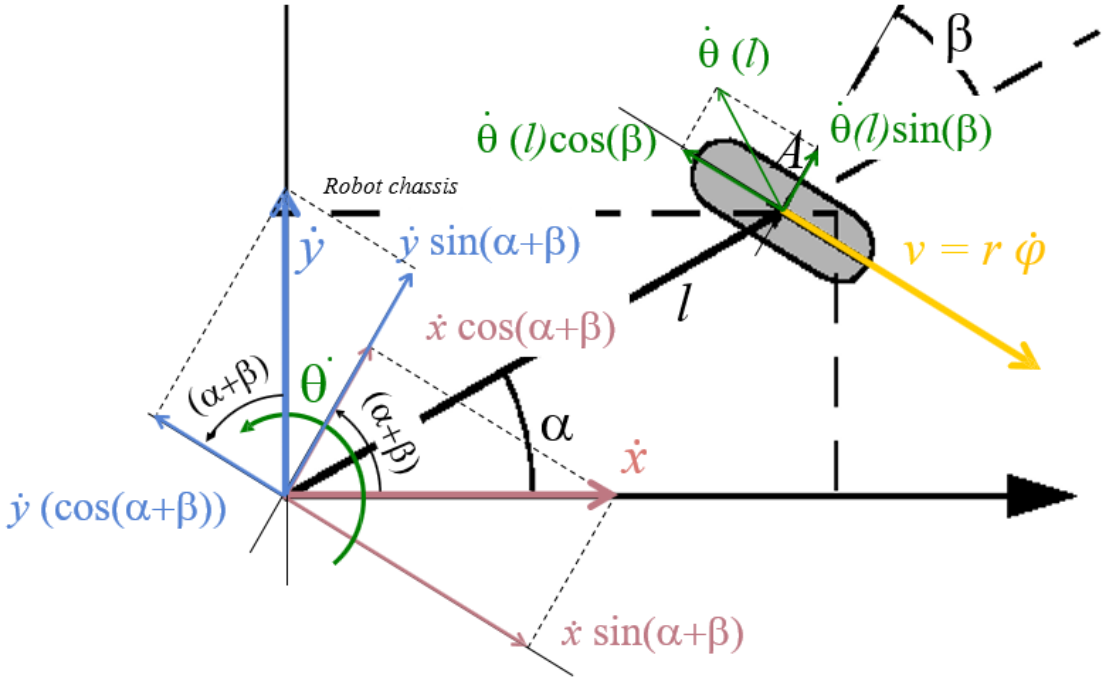


Fixed standard wheel

The sliding constraint: The component of the wheel's motion orthogonal to the wheel plane must be zero.

$$[\cos(\alpha+\beta)\ \sin(\alpha+\beta)\ \ l\sin\beta\,]\ \mathrm{R}(\theta)\dot{\xi}_I = 0$$

$$\dot{\xi}_R = \begin{bmatrix} \dot{x}_R \\ \dot{y}_R \\ \dot{\theta}_R \end{bmatrix}$$

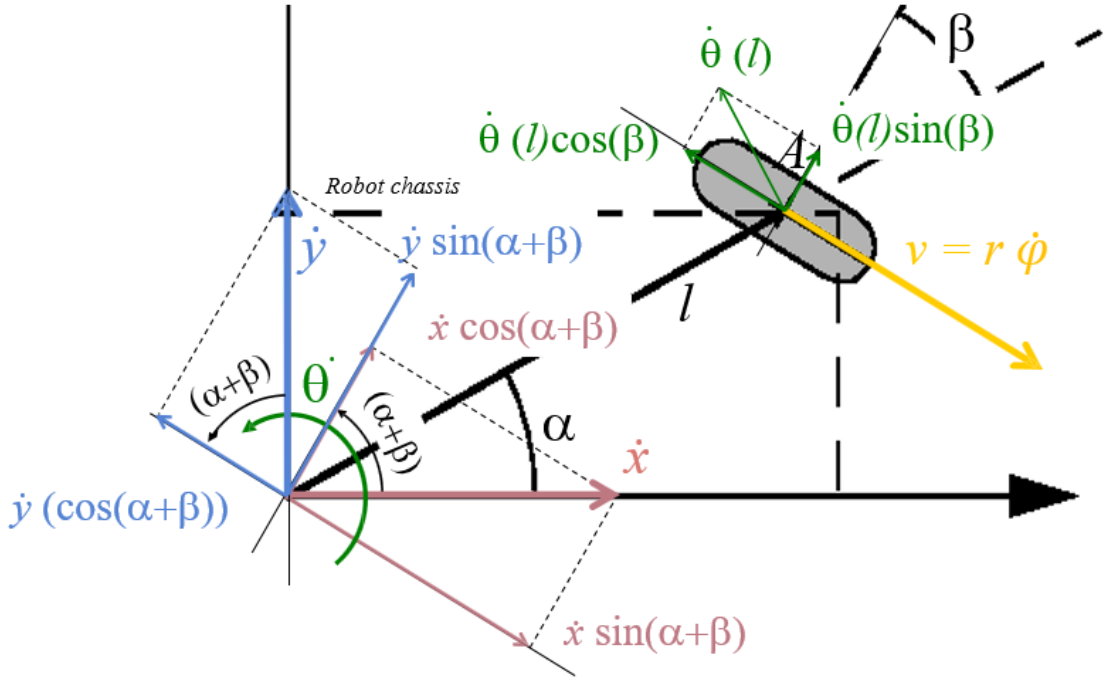


Steered standard wheel

The orientation of the wheel to the robot chassis is no longer a single value β, but instead varies as a function of time: $\beta(t)$.

$$[\sin(\alpha+\beta(t)) \; -\cos(\alpha+\beta(t)) \; (-l)\cos\beta(t)] \; \mathrm{R}(\theta)\dot{\xi}_I \; - \; r\dot{\varphi} = 0$$

$$[\cos(\alpha+\beta(t)) \; \sin(\alpha+\beta(t)) \; l \sin\beta(t)] \; \mathrm{R}(\theta)\dot{\xi}_I \; = 0$$



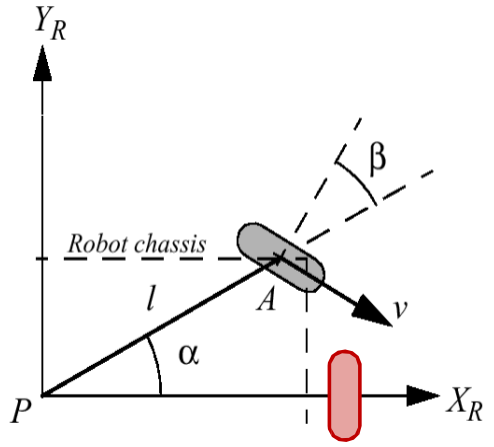
Example

Suppose that a fixed standard wheel *A* is in position such that $\alpha = 0$, $\beta = 0$ and $\theta = 0$. Please try to show the sliding constraint on this fixed standard wheel

$$[\cos(\alpha+\beta) \;\; \sin(\alpha+\beta) \; l \sin\beta \,] \; \mathrm{R}(\theta)\dot{\xi}_I = 0$$

- Solution: $\alpha=0$, $\beta=0$ would place the contact point of the wheel on X_I with the plane of the wheel oriented parallel to Y_I. If $\theta=0$, then the sliding constraint reduces to:

$$\begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = 0$$



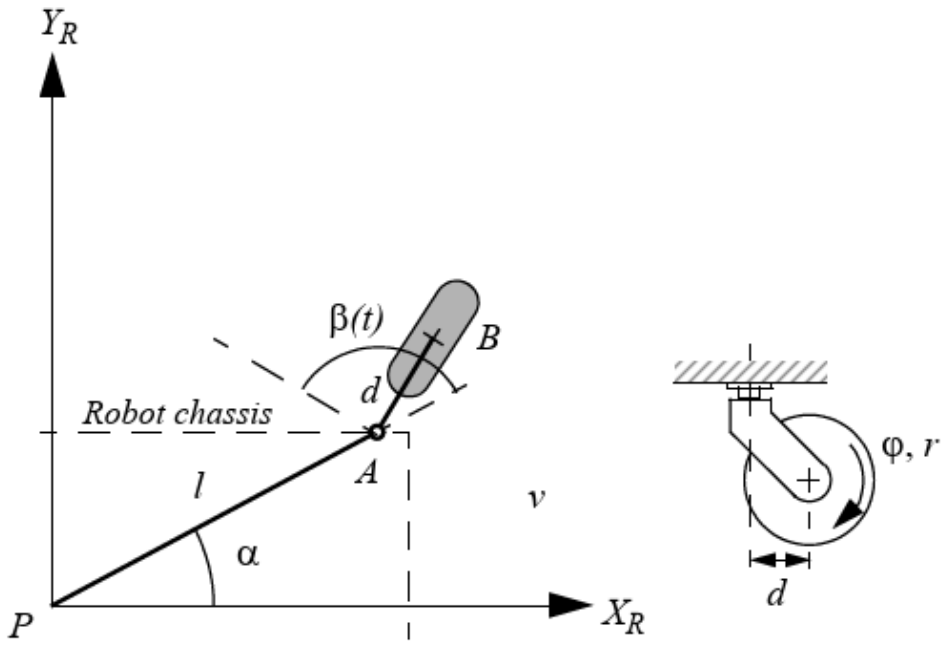
- This constrains the component along X_I to be zero, and since X_I and X_R are parallel for this case, the wheel is constrained from sliding sideways.

Castor wheel

No sliding constraint, but with an additional rolling constraint, in that appropriate rotation of the vertical axis must take place

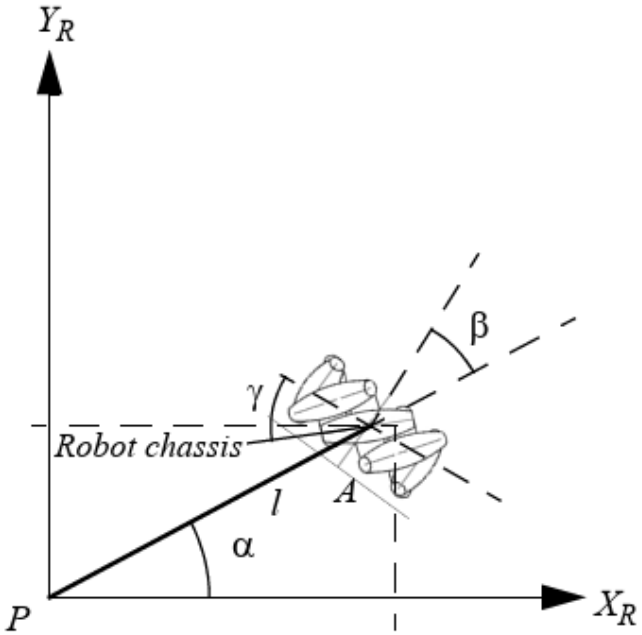
$$[\sin(\alpha+\beta) \;\; -\cos(\alpha+\beta) \;\; (-l)\cos\beta] \; R(\theta)\dot{\xi}_I \; - \; r\dot{\varphi} = 0$$

$$[\cos(\alpha+\beta) \;\; \sin(\alpha+\beta) \;\; l\sin\beta] \; R(\theta)\dot{\xi}_I + \; d\dot{\beta} = 0$$

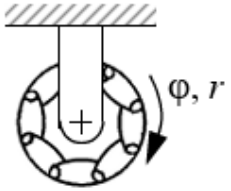


Swedish wheel

- γ is the angle between the roller axes and wheel plane. The instantaneous constraint is due to the specific orientation of the small rollers.
- The axis around which these rollers spin is a zero component of velocity at the contact point. That is , moving in that direction without spinning the main axis is not possible without sliding.



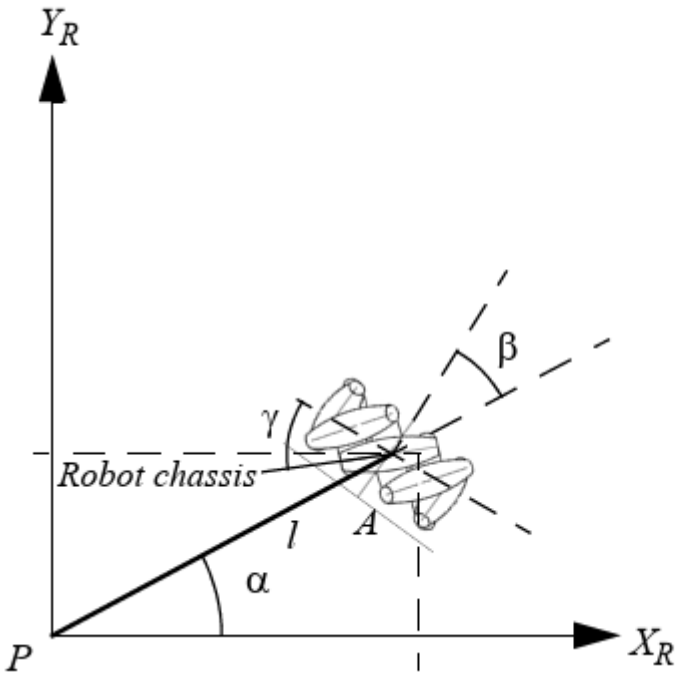




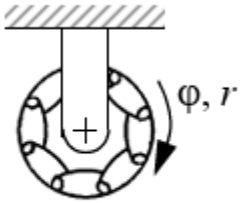
Swedish wheel

$$[\sin(\alpha+\beta+\gamma) \ -\cos(\alpha+\beta+\gamma)\ (-l)\cos(\beta+\gamma)]\ \mathrm{R}(\theta)\dot{\xi}_I \ - \ r\dot{\varphi}\cos\gamma = 0$$

$$[\cos(\alpha+\beta+\gamma) \ \sin(\alpha+\beta+\gamma)\ l\sin(\beta+\gamma)]\ \mathrm{R}(\theta)\dot{\xi}_I - \ r\dot{\varphi}\sin\gamma - r_{sw}\dot{\varphi}_{sw} = 0$$







Kinematic constraints: Complete robot

- Given a robot with M wheels, each wheel imposes zero or more constraints on the robot motion, only fixed and steerable standard wheels impose kinematic constraints
- Suppose we have a total of $N = N_f + N_s$ standard wheels, N_f and N_s are the numbers of fixed and steerable standard wheels. We can develop the equations for the constraints in matrix forms:

$$J_1(\beta_s)\mathrm{R}(\theta)\dot{\xi}_I - J_2\dot{\varphi} = 0 \qquad J_1(\beta_s) = \begin{bmatrix} J_{1f} \\ J_{1s}(\beta_s) \end{bmatrix} \qquad \varphi = \begin{bmatrix} \varphi_f(t) \\ \varphi_s(t) \end{bmatrix}$$

$$[\sin(\alpha+\beta) \ -\cos(\alpha+\beta) \ (-l)\cos\beta \,]\ \mathrm{R}(\theta)\dot{\xi}_I - r\dot{\varphi} = 0$$

$$[\sin(\alpha+\beta(t)) \ -\cos(\alpha+\beta(t)) \ (-l)\cos\beta(t)]\ \mathrm{R}(\theta)\dot{\xi}_I - r\dot{\varphi} = 0$$

Kinematic constraints: Complete robot

- Rolling constraints:

$$J_1(\beta_s)\mathrm{R}(\theta)\dot{\xi}_I - J_2\dot{\varphi} = 0 \qquad J_1(\beta_s) = \begin{bmatrix} J_{1f} \\ J_{1s}(\beta_s) \end{bmatrix} \qquad \varphi = \begin{bmatrix} \varphi_f(t) \\ \varphi_s(t) \end{bmatrix}$$

- Sliding constraints:

$$C_1(\beta_s)\mathrm{R}(\theta)\dot{\xi}_I = 0 \qquad C_1(\beta_s) = \begin{bmatrix} C_{1f} \\ C_{1s}(\beta_s) \end{bmatrix}$$

$$[\cos(\alpha+\beta) \;\; \sin(\alpha+\beta) \;\; l\sin\beta]\;\mathrm{R}(\theta)\dot{\xi}_I = 0$$

$$[\cos(\alpha+\beta(t)) \;\; \sin(\alpha+\beta(t)) \;\; l\sin\beta(t)]\;\mathrm{R}(\theta)\dot{\xi}_I = 0$$

Mobile robot maneuverability

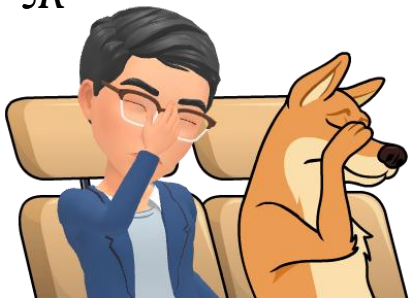
- The maneuverability of a mobile robot is the combination of the **mobility available** based on the sliding constraints plus additional freedom contributed by the steering
- The overall maneuverability of a robot is a combination of the **mobility available** based on the kinematic sliding constraints of the standard wheels, plus the **additional freedom** contributed by steering and spinning the steerable standard wheels.
- It can be derived:
  - Degree of mobility δ_m
  - Degree of steerability δ_s
  - **Robot maneuverability = available mobility + steerability**

$$\delta_M = \delta_s + \delta_m$$



Degree of mobility

- avoid any lateral slip the motion vector $\dot{\xi}_R$ has to satisfy the following constraints:



$$C_{1f}\mathrm{R}(\theta)\dot{\xi}_I = 0$$

$$C_{1s}(\beta_s)\mathrm{R}(\theta)\dot{\xi}_I = C_{1s}(\beta_s)\,\dot{\xi}_R = 0$$

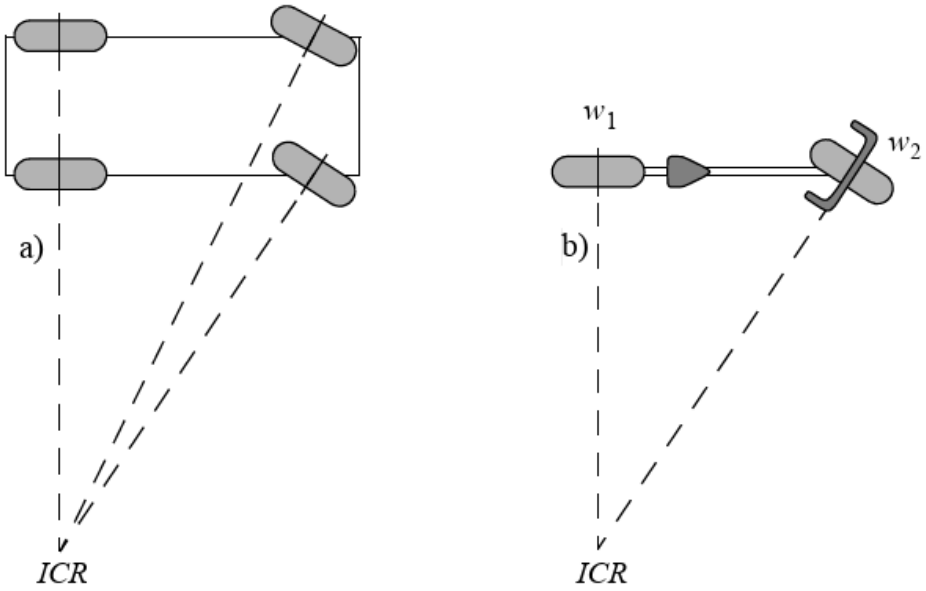
- Based on linear algebra, must belong to the **null space** of the projection matrix $C_{1s}(\beta_s)$.
- The **null space** of the matrix $C_{1s}(\beta_s)$ is the space N such that for any vector x in N

$$C_{1s}(\beta_s)\,x = \mathbf{0}$$

- Geometrically this can be shown by the **Instantaneous Center of Rotation** (ICR)

Instantaneous center of rotation

- The kinematic constraints can be demonstrated geometrically using the concept of a robot's ***Instantaneous Center of Rotation*** (ICR)
- The wheel must be moving instantaneously along some circle of radius *R* such that the center of that circle is located on the zero motion line. This center point, called ICR, may lie anywhere along the zero motion line.



Degree of mobility

- Robot chassis kinematics is a function of the set of independent constraints $\mathrm{rank}(\mathrm{C}_1(\beta_s))$

$$\mathrm{C}_1(\beta_s) = \begin{bmatrix} \mathrm{C}_{1f} \\ \mathrm{C}_{1s}(\beta_s) \end{bmatrix} \qquad \begin{aligned} C_{1f}\mathrm{R}(\theta)\dot{\xi}_I &= 0 \\ C_{1s}(\beta_s)\mathrm{R}(\theta)\dot{\xi}_I &= 0 \end{aligned}$$

- The greater the rank of $\mathrm{C}_1(\beta_s)$ the more constrained is the mobility

$$\delta_m = \mathrm{dim}\mathrm{N}(\mathrm{C}_1(\beta_s)) = 3 - \mathrm{rank}(\mathrm{C}_1(\beta_s))$$

- Mathematically
  - No standard wheels: $\mathrm{rank}(\mathrm{C}_1(\beta_s)) = 0$
  - All direction constrained: $\mathrm{rank}(\mathrm{C}_1(\beta_s)) = 3$
- Examples:
  - Unicycle: One single fixed standard wheel $\delta_m = 3 - \mathrm{rank}(\mathrm{C}_1(\beta_s))$=3-1=2
  - Differential drive: Two fixed standard wheels $\delta_m = 3 - \mathrm{rank}(\mathrm{C}_1(\beta_s))$=3-1=2

Degree of steerability

- Indirect degree of motion

$$\delta_s = \mathrm{rank}(C_{1s}(\beta_s))$$

$$C_{1f}\mathrm{R}(\theta)\dot{\xi}_I = 0$$

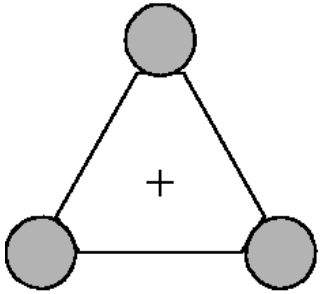
$$C_{1s}(\beta_s)\mathrm{R}(\theta)\dot{\xi}_I = 0$$

- The particular orientation at any instant imposes a kinematic constraint. However, the ability to change that orientation can lead additional degree of maneuverability
- Range of δ_s: $0 \leq \delta_s \leq 2$
- Examples
  - one steered wheel: Tricycle,
  - two steered wheels: No fixed standard wheel, pseudobicycle,
  - car (Ackermann steering): N_f = 2, N_s=2 with common axle,

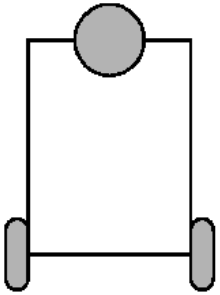
Robot maneuverability

- Degree of maneuverability: $\delta_M = \delta_s + \delta_m$
- Two robots with same δ_M are not necessary equal
  - Example: Differential drive and Tricycle
  - Their mobility and steerability could be different
- For any robot with $\delta_M = 2$ the ICR is always constrained to lie on a line
- For any robot with $\delta_M = 3$ the ICR is not constrained and can be set to any point on the plane

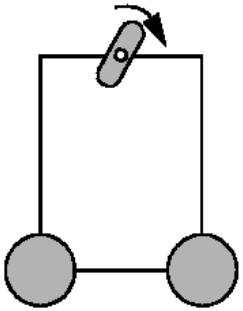
Basic types of 3 wheel configuration



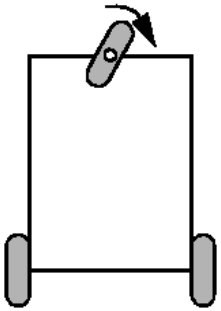
Omnidirectional
$\delta_M = 3$
$\delta_m = 3$
$\delta_s = 0$



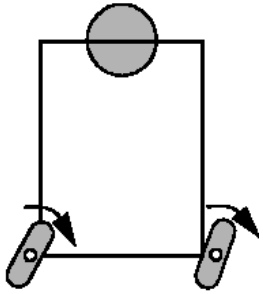
Differential
$\delta_M = 2$
$\delta_m = 2$
$\delta_s = 0$



Omni-Steer
$\delta_M = 3$
$\delta_m = 2$
$\delta_s = 1$

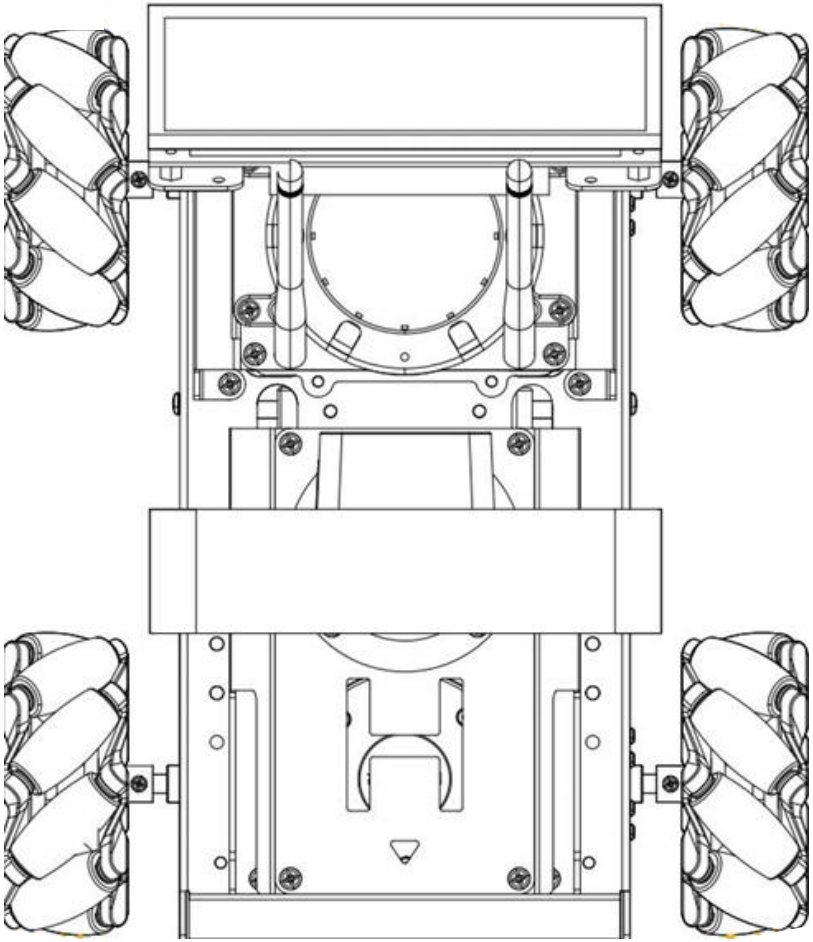


Tricycle
$\delta_M = 2$
$\delta_m = 1$
$\delta_s = 1$



Two-Steer
$\delta_M = 3$
$\delta_m = 1$
$\delta_s = 2$

Kinematics of Mecanum wheel car





Mecanum wheel car

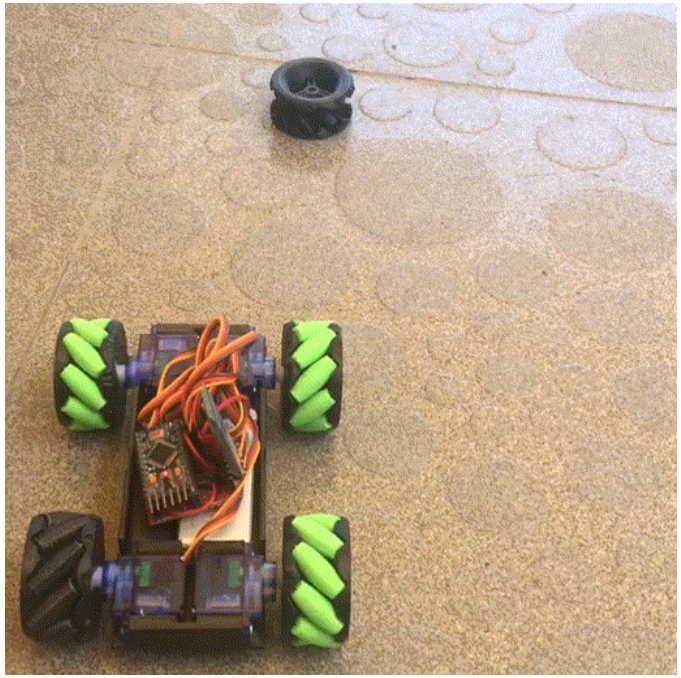


- A **Mecanum wheel** consists of a central hub and multiple barrel-shaped rollers installed around the rim.
- The axis of each roller is set at a 45° relative to the wheel's main rotation axis. Four wheels are used as a set: two left-handed (A wheels) and two right-handed (B wheels), which are mirror-symmetric.
- Common assembly patterns include AAAA, BBBB, AABB, ABAB, and BABA. Not all patterns support full motions such as forward/backward, rotation in place, and lateral left/right translation. The chassis of our car in ROSE5730 uses the **ABAB** pattern, which enables omnidirectional movement.

Physical characteriastics

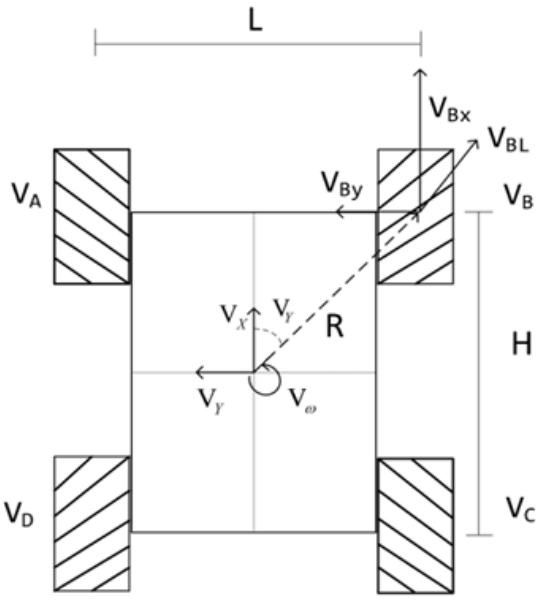
- By appropriately coordinating wheel directions and speeds, the vector sum of the individual wheel forces produces a resultant force in any commanded direction, allowing the platform to move freely **without changing** wheel steering angles.

- Because the rollers are arranged obliquely along the rim, the wheel can generate a lateral component and hence **slide sideways**. When the wheel rotates about its main axle, the envelope traced by the rollers forms a cylindrical surface, so the wheel can roll continuously.



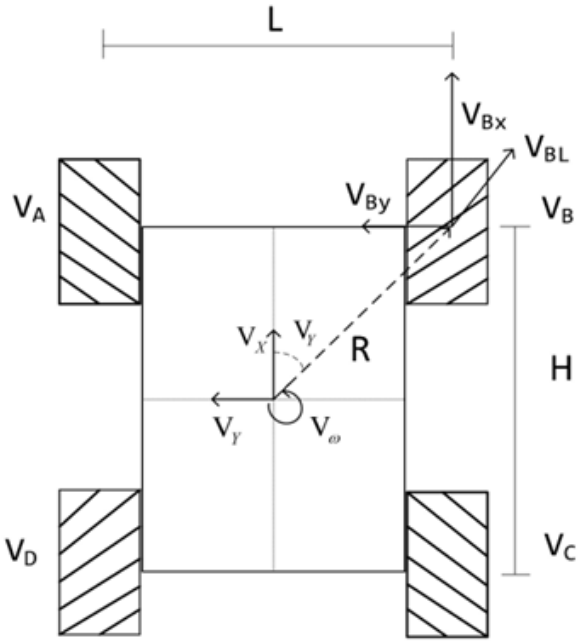
Variables and parameters

- V_X : chassis linear speed along the X-axis (forward/backward).
- V_Y : chassis linear speed along the Y-axis (left/right).
- ω: chassis angular speed about the vertical Z-axis (yaw).
- V_A, V_B , V_C , V_D : linear (tangential) speeds at the four wheels (A: front-left, B: front-right, C: rear-left, D: rear-right).
- L : distance between the left and right wheel centers (track width).
- H : distance between the front and rear wheel centers (wheelbase).
- θ : angle between the chassis center–to–front-right wheel line and the chassis axes (45°).



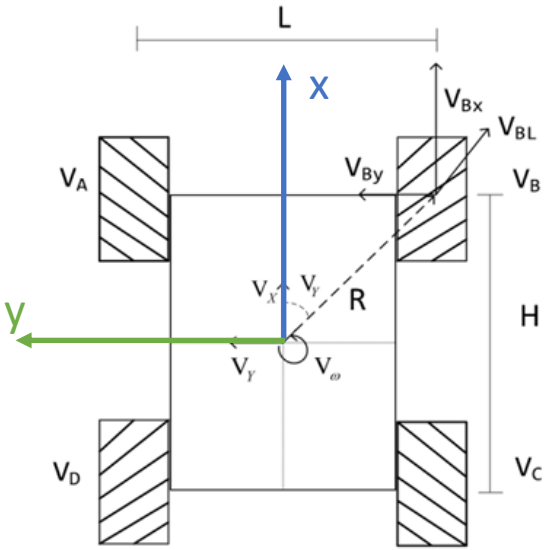
Assumptions for kinematics analysis

- Assumptions
  - The omni/Mecanum wheels do not slip relative to the ground, and ground friction is sufficient.
  - The four wheels are located at the four corners of a rectangle (or square), and opposing wheels are parallel.



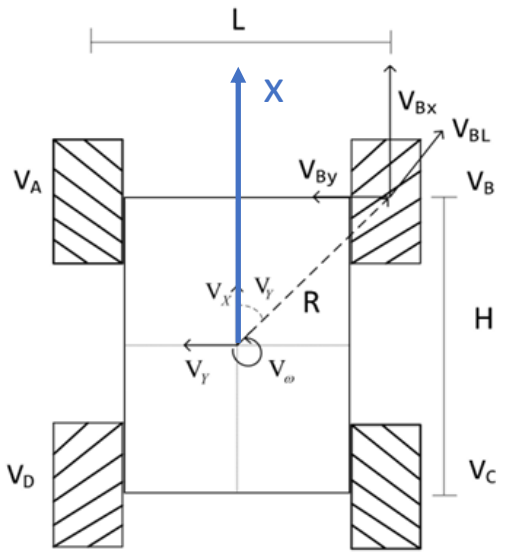
Coordinate Frames and Parameters

- Define X forward, Y left, Z up (right-handed frame).
- V_X : linear speed along X; V_Y : linear speed along Y; ω: angular (yaw) speed about Z (CCW positive).
- Wheel labels: A (front-left), B (front-right C (rear-left), D (rear-right); ABAB arrangement.
- L: track width (left–right wheel-center distance); H: wheelbase (front–rear wheel-center distance).
- Let r = (L/2 + H/2) be the effective moment arm from chassis center to each wheel contact point.



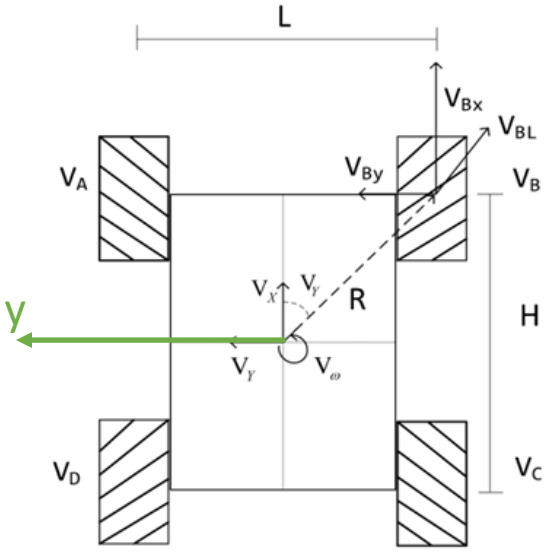
Pure translation along X-axis

- When the chassis translates purely along +X with speed V_X ($V_Y = 0$, $\omega = 0$), each wheel contributes the same forward tangential speed because the roller-induced lateral component cancels within the symmetric ABAB layout:
  - $V_A(x) = + V_X$
  - $V_B(x) = + V_X$
  - $V_C(x) = + V_X$
  - $V_D(x) = + V_X$



Pure translation along Y-axis

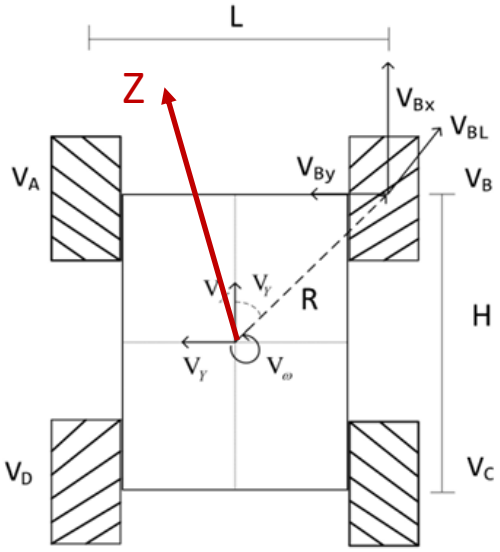
- For pure lateral motion along +Y with speed $V_Y(V_X=0, \omega=0)$, the 45° rollers project the wheel-ground force to produce opposite signs on the left and right sides under the ABAB configuration:
  - $V_A(y) = -V_Y$
  - $V_B(y) = +V_Y$
  - $V_C(y) = -V_Y$
  - $V_D(y) = +V_Y$



Pure rotation about Z-axis

- For pure yaw rotation at rate ω ($V_X = 0$, $V_Y = 0$), the tangential speed required at each wheel is proportional to the moment arm r. With counterclockwise $\omega > 0$ and the sign convention above:
  - $V_A(\omega) = -r \cdot \omega$
  - $V_B(\omega) = +r \cdot \omega$
  - $V_C(\omega) = -r \cdot \omega$
  - $V_D(\omega) = +r \cdot \omega$

  Here, $r = (L/2 + H/2)$.

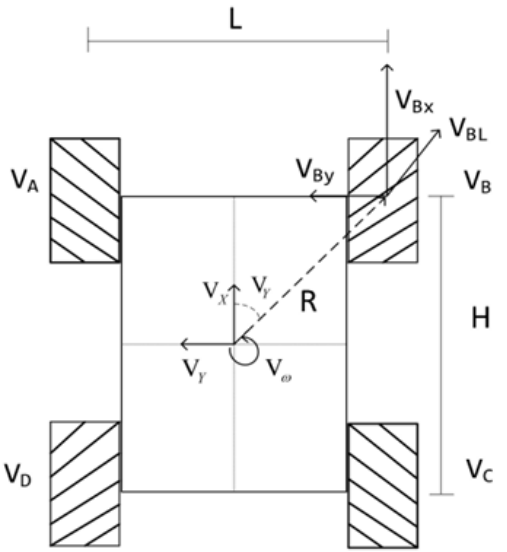


Combined motion

- For a general motion (V_X , V_Y, ω), superpose the three components: $V_i = V_i(x) + V_i(y) + V_i(\omega)$. The total wheel tangential speeds are:
  - $V_A = V_X - V_Y - r \cdot \omega$
  - $V_B = V_X + V_Y + r \cdot \omega$
  - $V_C = V_X - V_Y - r \cdot \omega$
  - $V_D = V_X + V_Y + r \cdot \omega$

  where $r = (L/2 + H/2)$.

$$\begin{bmatrix} V_A \\ V_B \\ V_C \\ V_D \end{bmatrix} = \begin{bmatrix} 1 & -1 & -r \\ 1 & 1 & r \\ 1 & -1 & -r \\ 1 & 1 & r \end{bmatrix} \begin{bmatrix} V_x \\ V_y \\ \omega \end{bmatrix}$$



ROSE5730 Autonomous Mobile Robots

Lecture 3: Kinematics

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